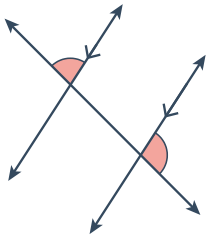
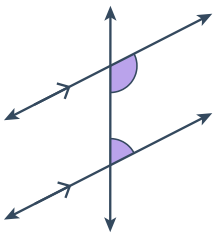
Parallel lines and transversals

1 Identify the types of angles formed by the transveral in the following diagrams:

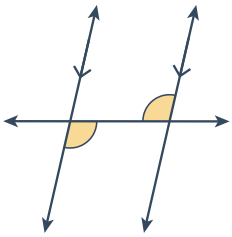
a



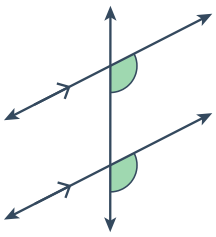
b



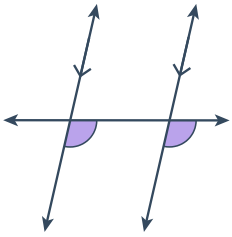
c



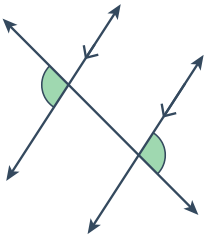
d



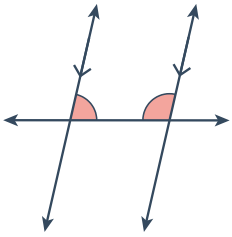
e



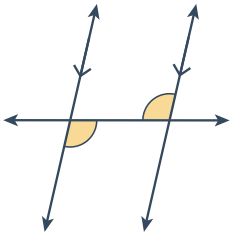
f



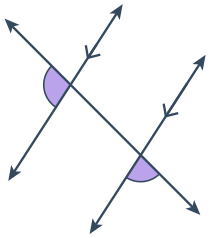
g



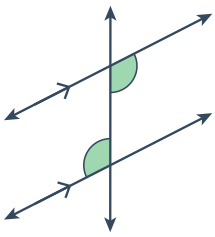
h



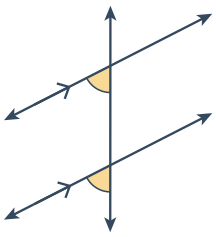
i



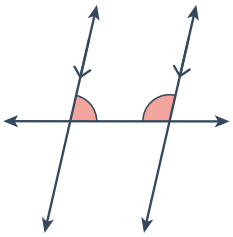
j



k



l

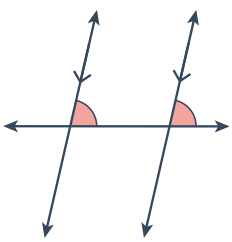


2 Consider the following diagrams:

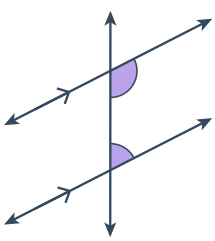
i State whether the two highlighted angles are congruent, complementary or supplementary.

ii Identify the types of angles formed by the transveral.

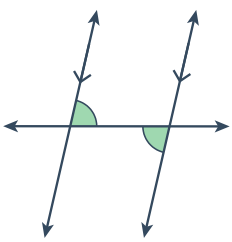
a



b

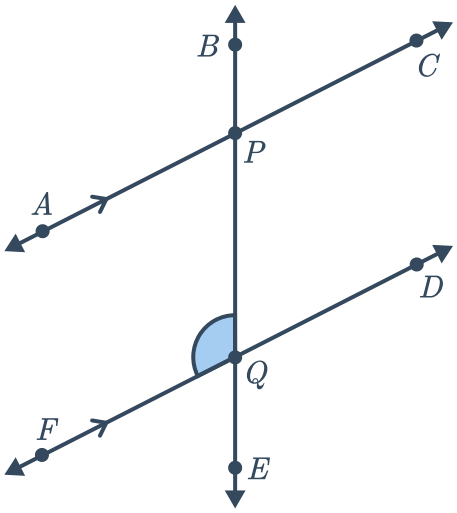


c

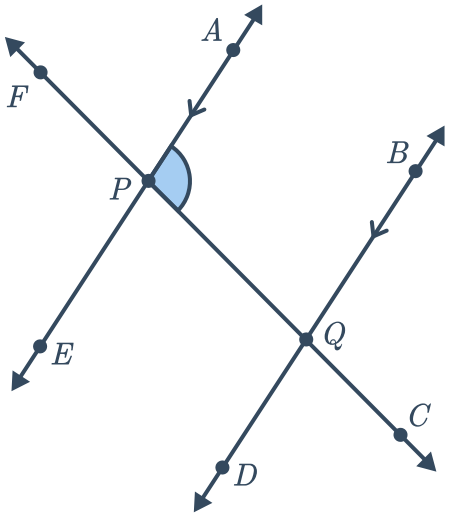


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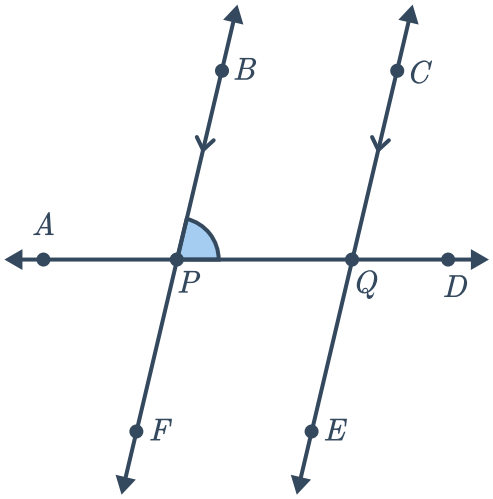
3 Consider the given diagram:
Write the angle that is corresponding to $\angle BQF$.



4 Consider the given diagram:
Write the angle that is alternate interior to $\angle APC$.



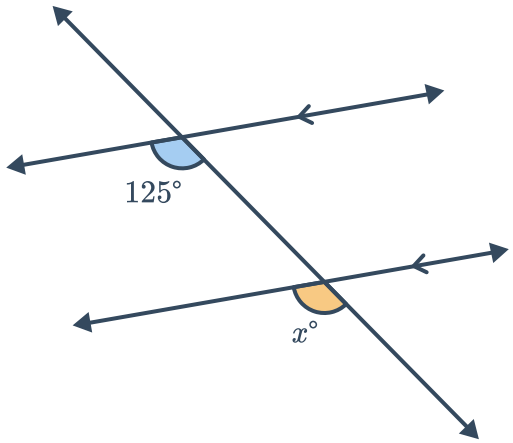
5 Consider the given diagram:
Write the angle that is consecutive interior to $\angle BPD$.



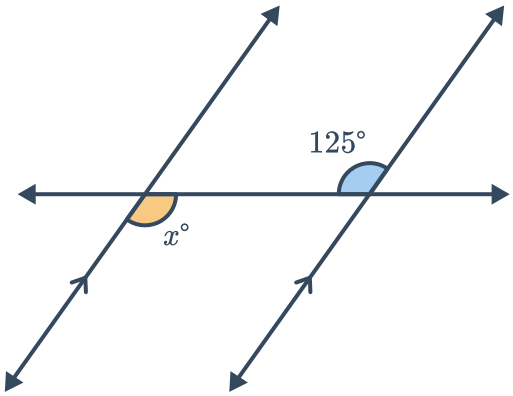
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6 For the following, solve for the value of x.

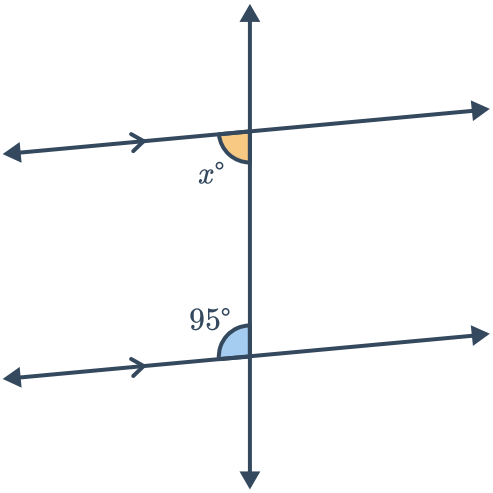
a



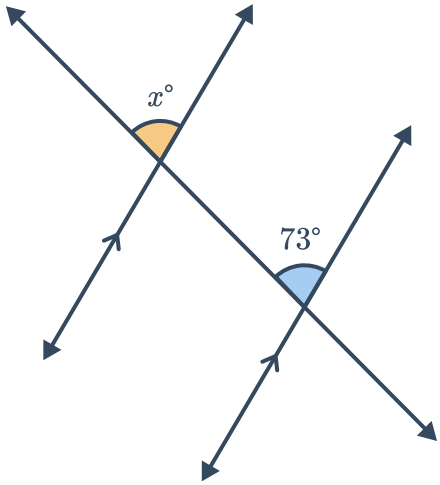
b



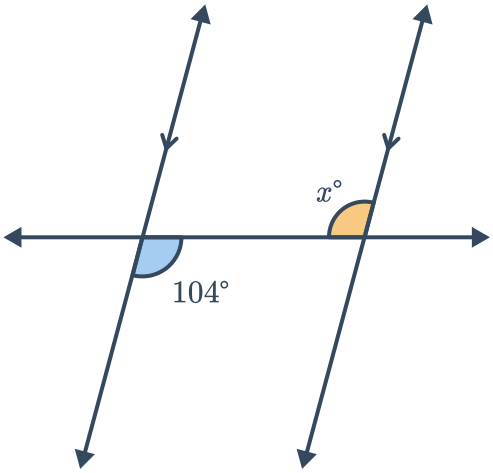
c



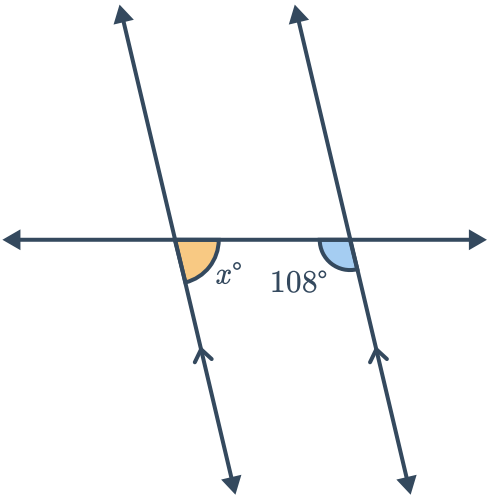
d



e



f

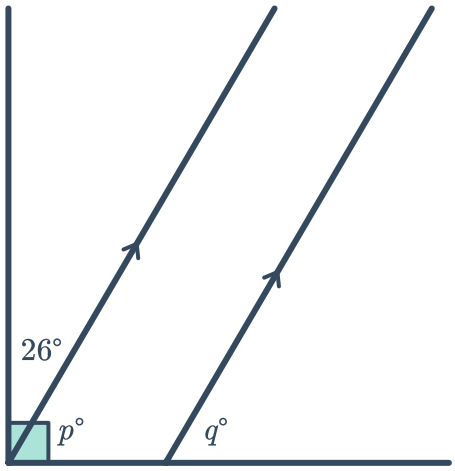


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7 Consider the given diagram:

a Solve for p.

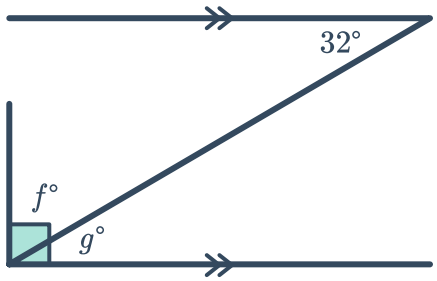
b Solve for q.



8 Consider the given diagram:

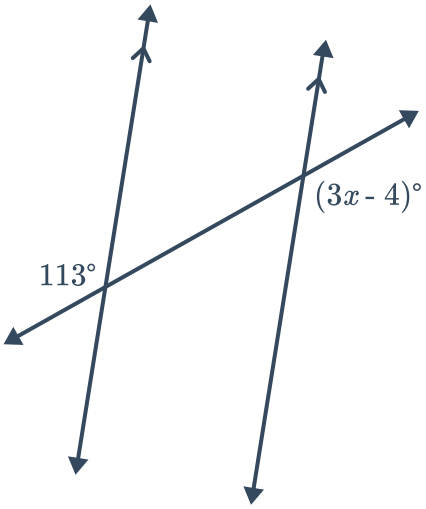
a Solve for g.

b Solve for f.

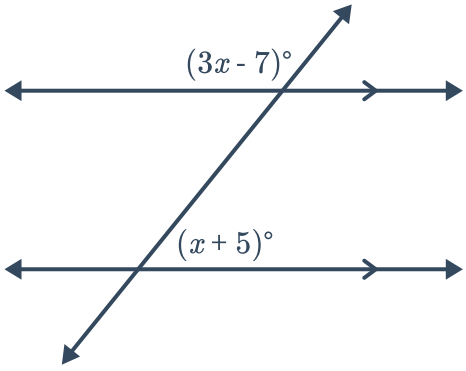


9 For the following, solve for x.

a

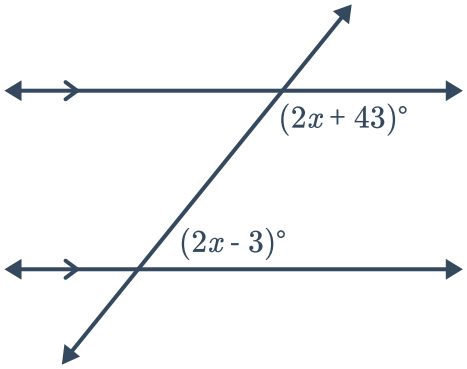


b

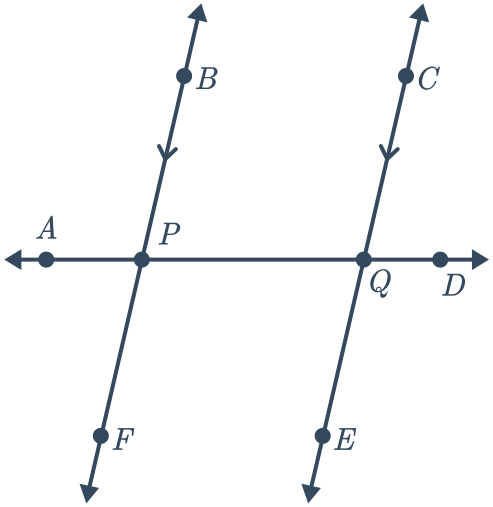


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c



10 Consider the given diagram:

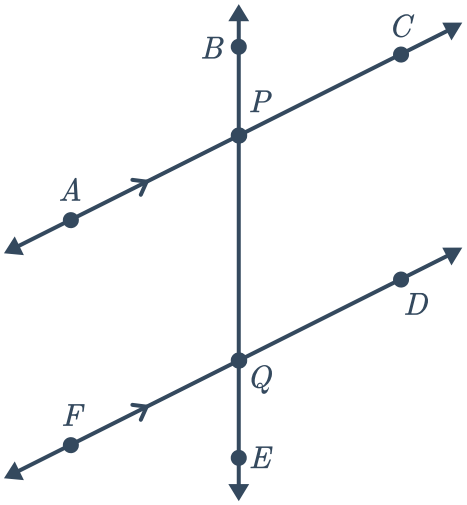


a How are $\angle DPF$ and $\angle AQC$ related?

b How are $\angle AQC$ and $\angle CQD$ related?

c If $\angle DPF = 102^\circ$, find $m\angle CQD$.

11 Consider the given diagram:



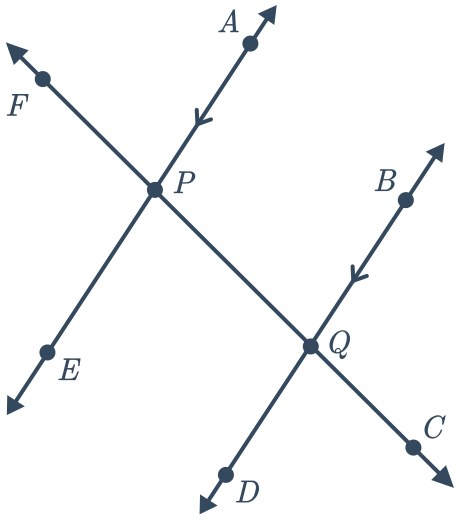
a How are $\angle APB$ and $\angle CPE$ related?

b How are $\angle CPE$ and $\angle BQD$ related?

c If $m\angle APB = 106^\circ$, find $m\angle BQD$.

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12 Consider the given diagram:



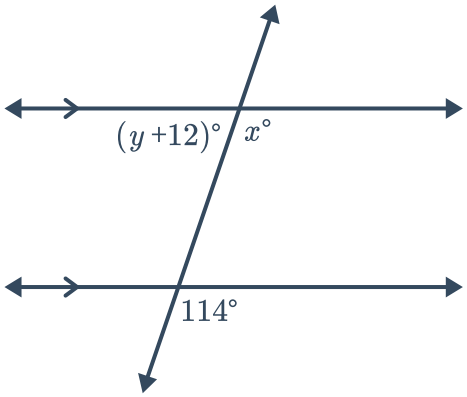
a How are $\angle DQF$ and $\angle CPE$ related?

b How are $\angle CPE$ and $\angle EPF$ related?

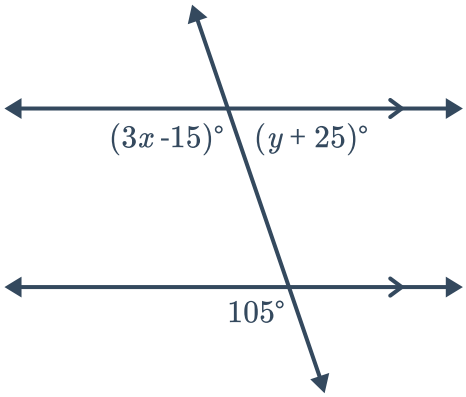
c If $m\angle DQF = 102^\circ$, find $m\angle EPF$.

13 For the following diagrams, solve for the value of x and y.

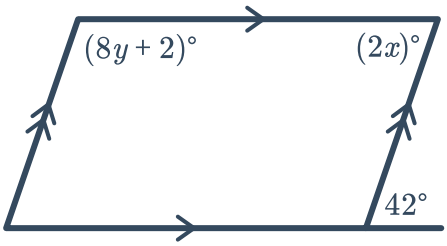
a



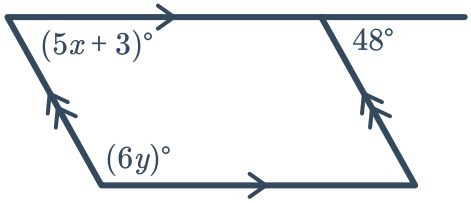
b



c



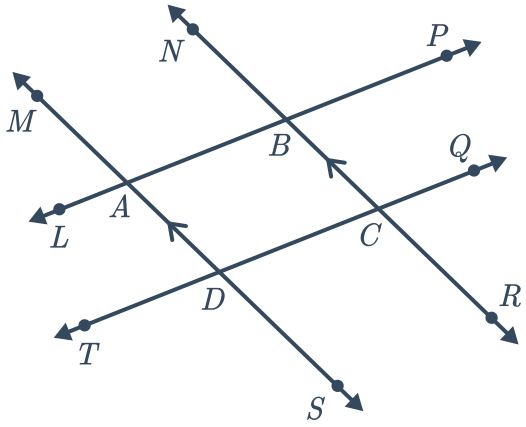
d



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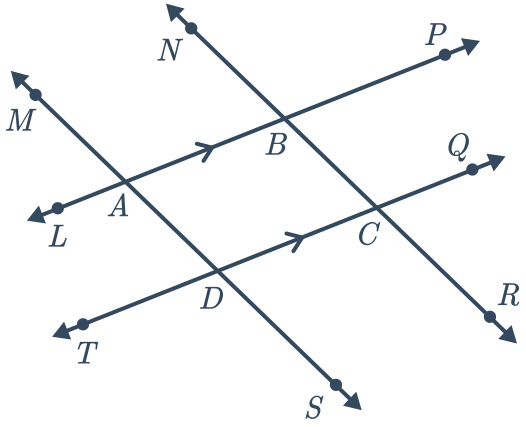
14 Consider the following diagram:
Given that $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$, which of the following are pairs of congruent angles?

- $\angle ADC$ and $\angle DCR$
- $\angle NBA$ and $\angle QCR$
- $\angle BAD$ and $\angle ADT$
- $\angle ABC$ and $\angle BAD$



15 Consider the following diagram:
Given that $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$, which of the following are pairs of congruent angles?

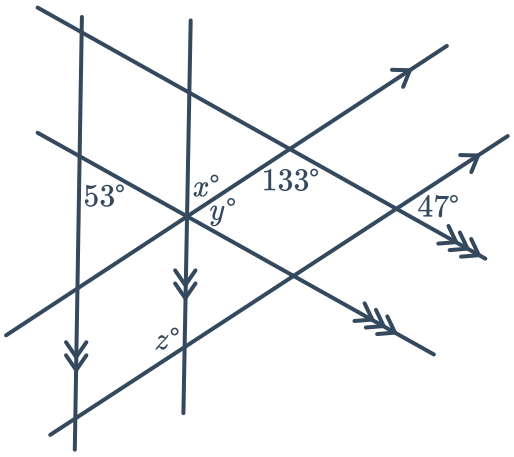
- $\angle NBP$ and $\angle DCR$
- $\angle BCQ$ and $\angle TDS$
- $\angle NBA$ and $\angle BAD$
- $\angle MAL$ and $\angle TDS$



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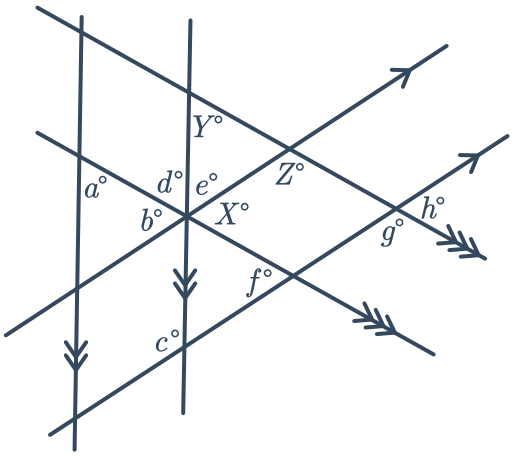
16 Consider the given diagram, which has three pairs of parallel lines. Find the value of the following:

a y **b** x **c** z



17 Consider the given diagram, which has three pairs of parallel lines. Write all variables that have the same value as the following:

a X **b** Y **c** Z



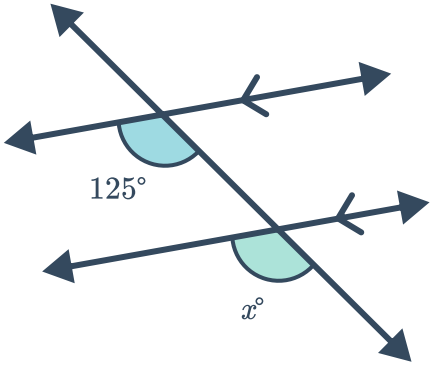
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Additional questions

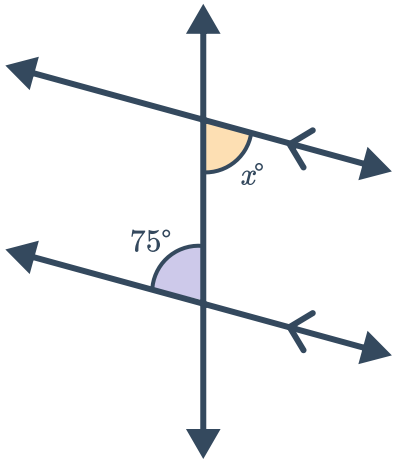
18 For each of the following diagrams:

- **i** Identify the types of angles formed by the transveral in the following diagrams
- **ii** Describe how the highlighted angles are related to each other
- **iii.** Find the value of x

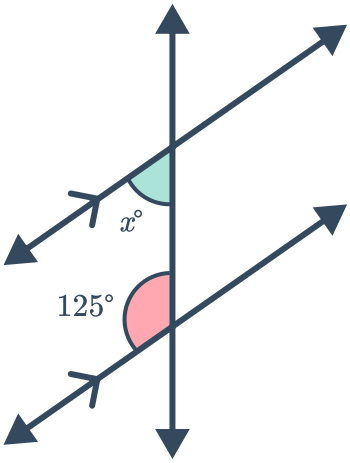
a



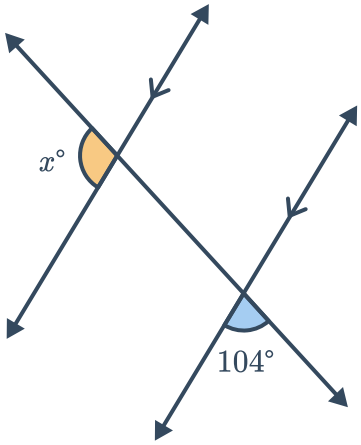
b



c

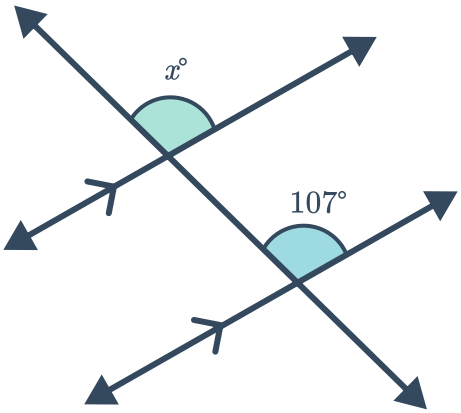


d

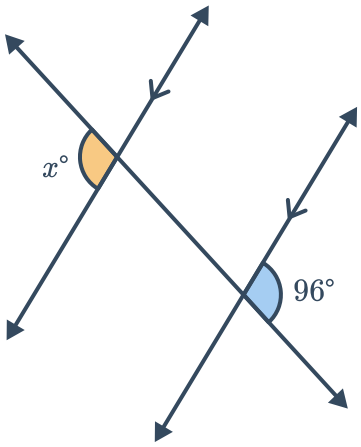


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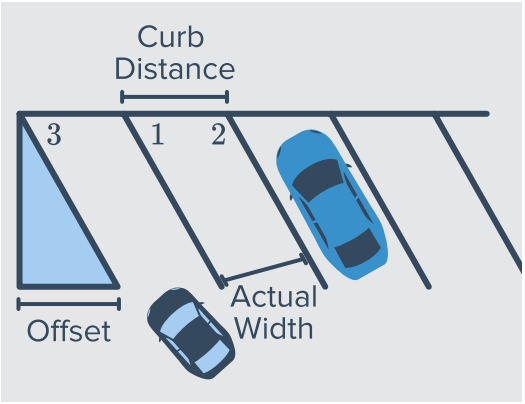
e



f

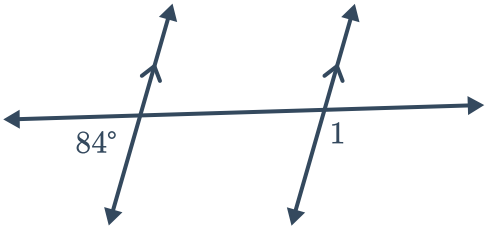


19 In the figure of the parking lot shown, the lines from the parking spaces should be parallel. If $m\angle 3 = 51^\circ$, what should $m\angle 1$ and $m\angle 2$ be? Explain your reasoning.



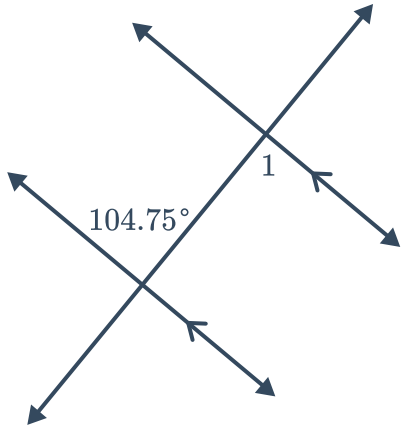
20 Review the following diagrams and statements. Determine whether the statements are correct. Explain your reasoning.

a $m\angle 1 = 84^\circ$ by the consecutive exterior angles theorem



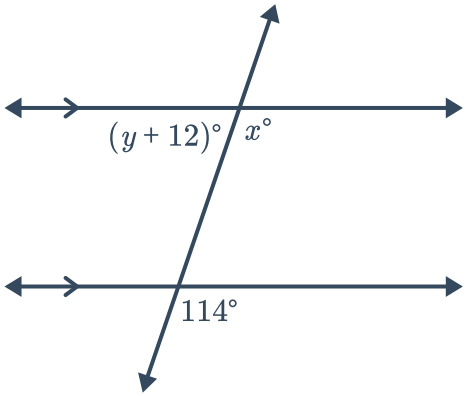
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b $m\angle 1 = 75.25^\circ$ by the alternate interior angles theorem

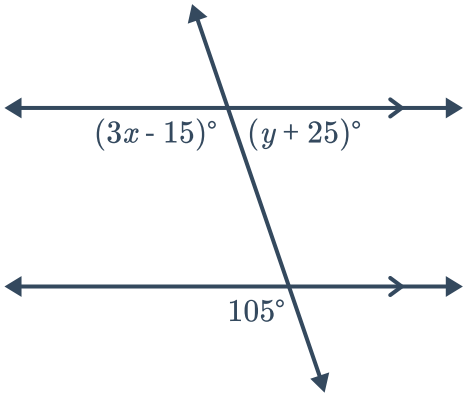


21 Consider the following diagrams:
Find the value of the unknown variable(s)

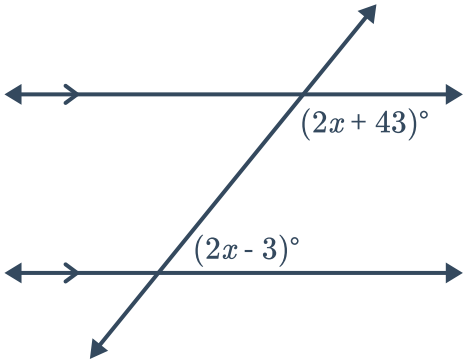
a



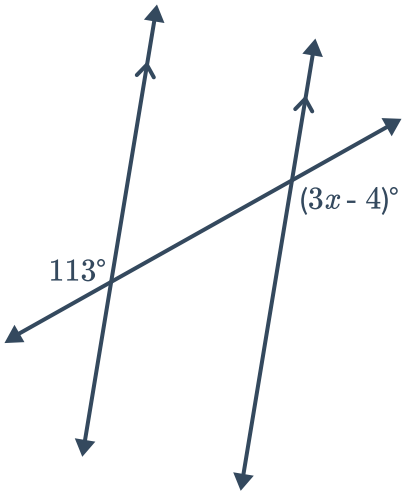
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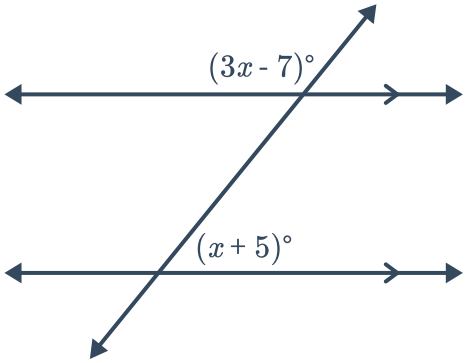


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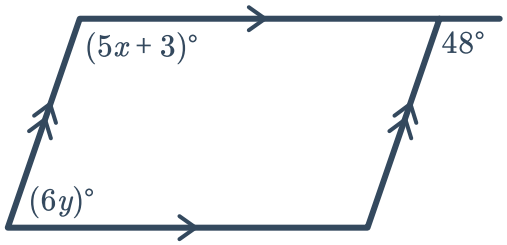


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e



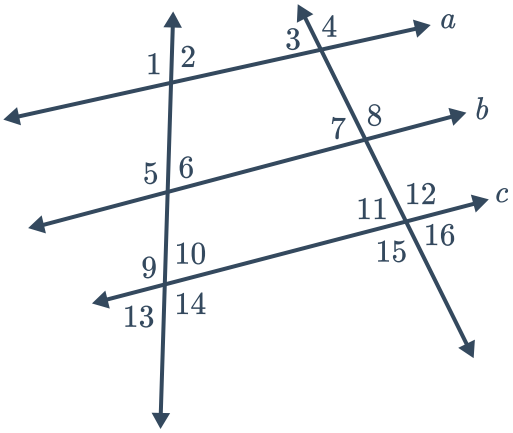
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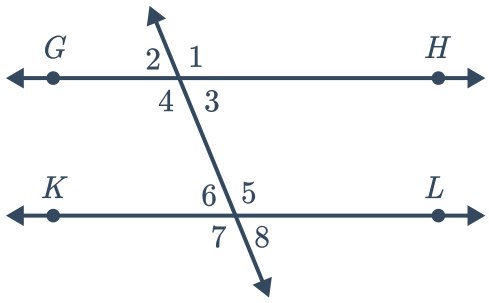
22 Determine if the information given is enough to justify the conclusion.

a Given: $a \parallel b$ and $\angle 1 \cong \angle 9$
Conclusion: $\angle 5 \cong \angle 9$

b Given:$\angle MNO$ and $\angle ONP$ are corresponding angles
Conclusion: $\angle MNO \cong \angle ONP$



23 Prove the alternate exterior angles theorem:
Given: $\overleftrightarrow{GH} \parallel \overleftrightarrow{KL}$
Prove: $\angle 2 \cong \angle 8$

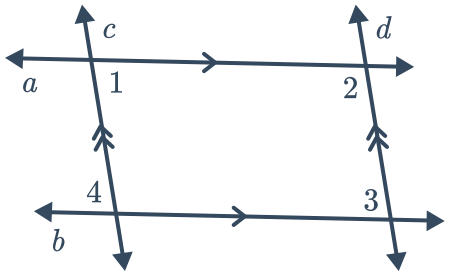


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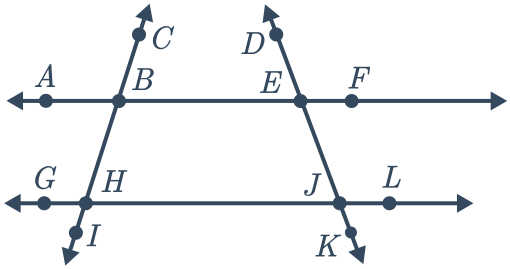
24

a Given: $a \parallel b, c \parallel d$
Prove: $\angle 1 \cong \angle 3$

b Prove part (a) in a different way.



25 Given: $\overleftrightarrow{AF} \parallel \overleftrightarrow{GL}$ and $\angle BHJ \cong \angle HJE$
Prove: $\angle HBE \cong \angle BEJ$



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